# Minimal block IV: the mixed-block envelope and the freezing inequalities

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**Abstract**

Two inputs for the continuity freezing argument. If one coordinate of a $(d_0+2)$-block carries an exponent above $p$ while the other $d_0+1$ carry exactly $p$, then $c^p Z(c) \le K(1+\log_+(cB))^{d_0}$ at every scale: one logarithm is lost. In box form, inserting a factor $u_i$ into an equal-exponent block produces such a mixed block after relabelling by the transposition $(0\,i)$ (the Hölder-insertion estimate). The mean-value bound $|\eta_1 f_{\xi_1}(s) - \eta_2 f_{\xi_2}(s)| \le (|\eta_1-\eta_2| + Me^{\beta/2}|\xi_1-\xi_2|)e^{-(\beta/2)s^2+\beta Ls}$ and the corner/strip split $D \le \omega + (D_{\max}/\delta)\sum_i u_i$ complete the toolkit. Zero `sorry`/`axiom`.

## 1 The things formalised

```
theorem iterChartGen_mixed_le ... (hp :   l, 1   l → l   d  + 1 → chartExp k h l = p)
    (hlt : p < chartExp k h 0) :
      K :  , 0   K    c, 0 < c → c ^ p * iterChartGen   a b k h (d  + 2) c
        K * (1 + logPlus (c * b ^ (  i   Finset.range (d  + 2), k i))) ^ d
def shiftExp {m :  } (h : Fin m →  ) (i : Fin m) : Fin m →   :=
  fun j => h j + if j = i then 1 else 0
theorem boxIntegralFin_shift_le ... (hp :   j, finExp k h j = p) (i : Fin (d  + 2)) :
      K :  , 0   K    c, 0 < c → c ^ p * boxIntegralFin   a b c k (shiftExp h i)
        K * (1 + logPlus (c * b ^ (  j, k j))) ^ d
theorem weighted_quadKernel_sub_le (  L M         s :  ) (h  : 0 <  ) (hs : 0   s)
    (h   : |  |   L) (h   : |  |   L) (h   : |  |   M) :
    |  * quadKernel     s -   * quadKernel     s|
        (|  -   | + M * Real.exp (  / 2) * |  -   |) * quadKernel (  / 2) (2 * L) s
theorem corner_strip_le {m :  } (u : Fin m →  ) (hu :   i, 0 < u i) (D    Dmax :  )
    (h  : 0 <  ) (h  : 0    ) (hDmax : 0   Dmax) (hcorner : (  i, u i    ) → D   )
    (hmax : D   Dmax) : D    + Dmax /   *   i, u i
```

## 2 Proof strategy

The mixed block is the exact tower $Z_{d_0+2}(c) = (\prod k_i^{-1}) D\, c^{-p} G_{d_0+2}(cB)$ with $G_{d_0+2} = \mathrm{iterDivPrim}\, G_2\, d_0$ and $G_2$ an admissible base (unit 57); the polynomial-log envelope of unit 74 bounds the tower at every argument. For the insertion estimate the transposition $\sigma = (0\,i)$ moves the shifted coordinate to index 0; the relabelled exponents are read off through `chartExp_toNatFun_eq_finExp`, and $\sum k$ is invariant by `Equiv.sum_comp`. The mean-value bound splits $\eta_1 e^{x_1} - \eta_2 e^{x_2} = (\eta_1-\eta_2)e^{x_1} + \eta_2(e^{x_1}-e^{x_2})$, uses $|e^x - e^y| \le |x-y|e^{\max(x,y)}$ (from $1+t \le e^t$), and absorbs the factor $\beta s$ by $\beta s\, e^{-(\beta/2)s^2} \le e^{\beta/2}$.

# 3 Roadblocks and resolutions

- The `wlog` symmetric case needs the two `abs_sub_comm` rewrites targeted (`abs_sub_comm (Real.exp y)`); the bare form rewrites the wrong occurrence.
- `push_neg` is deprecated on this pin: use `push Not`.
- Corner/strip: the nonnegativity of $\omega$ and $D_{\max}$ must be hypotheses; without them the strip case cannot absorb $\omega$.

# 4 Outcome

Astra target 3 (strip and Hölder-insertion estimates) is done in the form needed for the freezing argument. Next: the freezing theorem $Z - Z_0 = o(N^{-p} \log^d N)$ and the multiplicity-$m$ leading theorem for general continuous $\xi, \eta$.